

A Demonstration of ShiftingGolf: Gross Motor Skill Correction Using Redirection in VR

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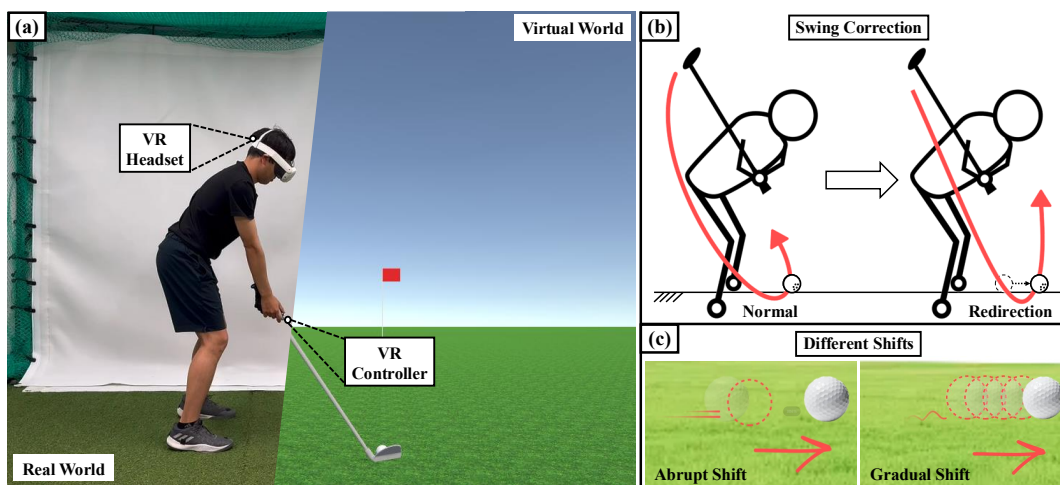


Figure 1: Our portable VR system allows users to practice golf swings with a physical club while experiencing real-time redirection in a virtual environment (a). The system subtly shifts the virtual ball just before impact to guide users toward better swing paths (b). Different redirection patterns provide varied adaptation experiences (c).

Abstract

In golf, ingrained swing path errors often persist despite repeated practice, posing a major challenge to skill development. This work presents a portable VR training system that uses real-time redirection to help users subconsciously correct their swing path. By shifting the virtual golf ball just before impact, our method prompts intuitive adjustments without explicit feedback. Three redirection modes are implemented in the proposed system to explore visuo-motor adaptation under different temporal patterns. This hands-on demonstration highlights how subtle, well-timed interventions in VR can promote lasting motor skill improvement through immersive, self-guided training.

CCS Concepts

• Human-centered computing → Virtual reality.

Keywords

Virtual Reality, Visual Illusion, Redirection, Golf Training

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1 Introduction

Unlearning ingrained habits is a fundamental challenge in sports training. In golf, poor swing mechanics, such as an outside-in swing path, often lead to slice shots and directional inconsistencies [Kawada 2008; Shan and Zhang 2011]. These habits are difficult to correct without expert feedback and typically persist even with repeated practice.

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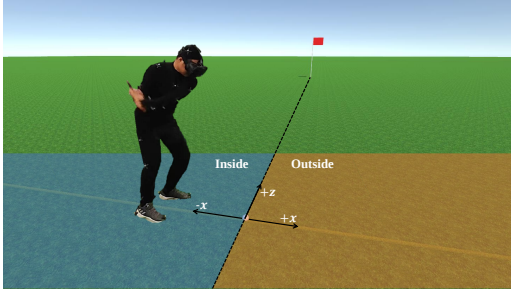


Figure 2: VR golf swing setup. The user faces along the yellow line and swings toward the red flag target in the virtual environment.

To address this, we present an interactive VR training system that subtly corrects swing path errors through real-time redirection. By shifting the virtual golf ball forward just before impact, the system prompts users to adjust their motion subconsciously, reinforcing a more optimal swing trajectory (Figure 1).

Building upon prior work [Liao et al. 2025] using motion capture in lab-based environments, this work proposes a demonstration showcasing a fully portable VR setup built with a standalone VR headset. Users can experience three distinct redirection modes: Abrupt, Gradual, and Interspersed, designed to guide swing correction without explicit instruction. Each mode explores different strategies for inducing visuomotor adaptation while preserving user immersion and comfort.

2 System Setup

This demonstration features a fully self-contained VR golf training system implemented on Meta Quest 3¹, designed for portability, ease of use, and real-time interaction without external sensors or markers. The system is developed using Unity², with VR content rendered and executed directly on the Quest 3 as a standalone device. Users hold a virtual golf club, visually represented in VR, and use a real controller in one hand to approximate swing position and orientation. This enables responsive swing detection and dynamic feedback. No external tracking devices, cables, or network infrastructure are needed.

As shown in Figure 2, the user performs golf swings while standing in a designated area. The VR scene simulates a driving range with a guide line and a visible target flag. When the swing is initiated, the system renders the ball's trajectory and redirection effect in real time. A new ball is automatically reset for the next swing, supporting continuous practice. This lightweight and mobile configuration allows the system to be used beyond lab settings, enabling training at home, in classrooms, or during public exhibitions.

3 Design of redirection

To support swing correction through subtle behavioral guidance, our system dynamically shifts the virtual golf ball just before impact. This redirection encourages users to instinctively adjust their swing path in response to the unexpected visual cue, without requiring explicit feedback or instruction.

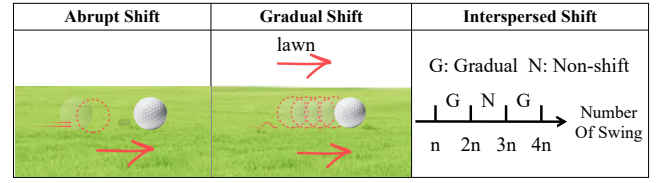


Figure 3: Ball shift modes. Abrupt shift: The ball instantly shifts two ball lengths. Gradual shift: The ball shifts the same distance gradually. The grass texture also moves to enhance imperceptibility. Interspersed shift: Alternates between gradual shift and no shift every n swings.

The ball begins to shift at the *transition from backswing to downswing*, a critical moment when the user is focused and committed to the motion. This timing leverages visuomotor adaptation principles, triggering subconscious corrections while preserving immersion. In this work, we implement and compare **three redirection modes**, as illustrated in Figure 3:

- **Abrupt Shift (A):** The ball shifts forward instantly by two ball lengths in a single frame.
- **Gradual Shift (G):** The ball shifts the same distance smoothly over 0.4 seconds, synchronized with typical downswing speed. To minimize detection, the grass texture is also shifted to create a consistent visual illusion.
- **Interspersed Shift (I):** A novel pattern that alternates between gradual shift and no shift every n swings, allowing users to consolidate corrected behavior while avoiding over-compensation.

These redirection patterns form the core of our demo experience, offering users a unique opportunity to feel how implicit visual cues can reshape physical performance in real-time. **More detailed information is provided in the previous paper** [Liao et al. 2025].

4 Conclusion

We present a VR-based golf training system that uses subtle visual redirection to help users correct swing path errors. By dynamically shifting the virtual ball just before impact, our system encourages subconscious motor adaptation without explicit feedback. Users experience three redirection modes: Abrupt, Gradual, and Interspersed, each designed to explore different strategies for inducing habit change. This demonstration highlights the feasibility of achieving effective redirection using a lightweight, portable setup with off-the-shelf hardware. The interspersed mode, in particular, balances long-term learning effects with perceptual subtlety, making it a promising direction for real-world skill training.

At **SIGGRAPH Asia Emerging Technologies**, attendees have the opportunity to interact directly with the system, experience each redirection mode, and observe how their motion adapts in real-time. This work demonstrates how redirection technologies can support intuitive, self-guided motor learning and opens the door to broader applications in physical training, rehabilitation, and beyond.

Acknowledgments

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¹<https://www.meta.com/jp/quest/quest-3/>

²<https://unity.com/>

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